

# Jadon Lai

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jadonlai.github.io/github-portfolio

## EDUCATION

**California Polytechnic State University, San Luis Obispo, CA**

2021 - 2024

- **Major:** B.S. in Computer Science, Concentration in AI and Machine Learning
- **GPA:** 3.5 | **Dean's List:** 4 quarters | **President's List:** 2023-2024

## SKILLS

- **AI:** Tensorflow, PyTorch, SciKit-Learn, CUDA, Data Science, Reinforcement Learning, Computational Neuroscience, Neural Networks, LLMs, Transformers, RAG, NLP, AI Agent Development, Prompt Engineering, Basic MCP
- **Frontend:** UI/UX design, React, React Native, Figma, TailwindCSS, Web/Mobile Development, Basic Swift
- **Computer Languages:** Python, TypeScript/JavaScript, C, HTML/CSS, SQL

## Karate

2009 - 2014, 2021 - 2023

- **Dragonstar Karate:** Trained for 6 years (5-6 days per week + independent practice) to earn a 1st Degree Black Belt
- **Shorin Ryu Karate:** Achieved Brown Belt (2nd kyu) through consistent, disciplined training and practice

## PROFESSIONAL EXPERIENCE

**AI/ML Engineer | MyBackHub**

Jan. 2026 - Present

- Design, train, and deploy models trained on 3M+ patient records to diagnose patients with back problems
- Build scalable data pipelines and feature engineering workflows to clean, transform, and analyze the large-scale healthcare dataset for predictive modeling

**Founder | Evros**

Jun. 2024 - Present

- Develop a cross-country management app using JavaScript, React Native, and Appwrite to streamline coach-to-athlete interactions
- Design intuitive UI/UX interfaces from scratch using Figma, prioritizing functionality and presentation

**Sports Coach | SLO High School**

Apr. 2023 - Present

- Coach cross country and track and field, focusing on safety, technique correction, and pacing
- Maintain active CPR and coaching certifications

**AI4SaR Instructional Student Assistant | Cal Poly SLO**

Jun. 2024 - Dec. 2024

- Spearheaded research and development of an AI-driven project locating missing persons using public transit
- Developed the Public Transit Search Widget, a predictive map tool indicating likely movements of missing persons
- Employed TensorFlow/Scikit-Learn to build 6 models: Decision Tree, Random Forest, ANN, RNN, LSTM, and BiLSTM
- Implemented Folium and Plotly for data analysis and geospatial mapping

## TECHNICAL PROJECTS

- **Document Intelligence:** AI Document Intelligence platform with a RAG-powered query-and-answer system
  - Architected infrastructure utilizing Docker and AWS for scalable containerization
  - Integrated a Qwen model, Supabase, and SQL with vector embeddings to optimize search latency and memory efficiency
  - Utilized Flask to connect to a React/Vite frontend with an interactive interface
- **LLM Research:** Produced 2 LLM's, using PyTorch and LLM architecture/features
  - Base LLM: 33 thousand parameter neural network using embeddings
  - Mock Llama: 2-million-parameter LLM, implementing RMSNorm pre-normalization, SwiGLU activation function, rotary position embeddings, multi-head attention, GPU/CUDA usage
- **Wikipedia Game Solver:** Implemented 4 AI algorithms (BFS, DFS, GBFS, and A\*) to solve the Wikipedia Game. Authored a comprehensive analytical report evaluating algorithmic performance and efficiency
- **Data Science For Cars:** Conducted graph analysis and data cleaning on a Kaggle dataset of used cars. Applied KNN and K-Means clustering to model vehicle suitability for students